Syllabus — Python Programming for Finance

Administrative information

Field	Value
Programme / Program	PGE
Campus	Multicampus (Lille / Sophia)
Year of study	M1
Academic year	2026/2027
Semester	Fall
Course title	Python Programming for Finance
Course leader	Alexandre LANDI
Language of instruction	English

Course description

This course introduces Python programming for financial data analysis. Students with **no prior programming experience** learn to write Python scripts, manipulate numerical data with NumPy, work with tabular financial datasets using pandas, import data from CSV and Excel files, and perform basic time-series operations.

The course uses finance-themed examples (stock prices, market listings, returns) but focuses on programming and data skills rather than financial theory.

Prerequisite: none. Familiarity with spreadsheets is helpful but not required.

Learning outcomes

Upon completion, students will be able to:

1. Write and debug short Python scripts using variables, built-in types, lists, and user-defined functions
2. Apply software engineering basics: readable code structure, docstrings, and simple testing concepts
3. Perform vectorized numerical operations with NumPy arrays
4. Parse and format dates with the `datetime` module; compare values across types
5. Create, explore, filter, and aggregate pandas DataFrames
6. Import and clean financial data from CSV, Excel, and online sources
7. Build and manipulate time-series indexes; apply rolling and expanding window operations
8. Read Python code and predict its output without executing it (exam skill)

DataCamp courses used

Midterm — exam covers:

- Introduction to Python for Finance
- Intermediate Python for Finance
- Data Manipulation with Pandas
- Importing and Managing Financial Data in Python
- Manipulating Time Series Data in Python

Final — exam covers all midterm courses, plus:

- Introduction to Functions in Python
- Software Engineering Principles in Python
- the in-class project

Pre-class work
Students complete the corresponding DataCamp course **before** each session. Class time is not used for first exposure to new material; it focuses on clarification, live coding, and exam-style exercises.

Schedule

Contact time: 12 sessions × 90 minutes (18 hours total).

Complete the listed DataCamp course **before** the corresponding session (see Pre-class work).

Session	Complete before class	In class
1	Introduction to Python for Finance	Python basics, lists, NumPy — clarification and exam-style exercises
2	Intermediate Python for Finance	Datetimes, comparisons, DataFrame basics — clarification and exam-style exercises
3	Data Manipulation with Pandas	DataFrames, filtering, aggregation, indexing — clarification and exam-style exercises
4	Importing and Managing Financial Data in Python	CSV/Excel import, cleaning, groupby — clarification and exam-style exercises
5	Manipulating Time Series Data in Python	Timestamps, resampling, rolling/expanding windows — clarification and exam-style exercises
—	**Midterm exam**	Covers the five courses above
6	Introduction to Functions in Python	User-defined functions, scope, docstrings — clarification and exam-style exercises
7	Software Engineering Principles in Python	Modularity, PEP 8, testing — clarification and exam-style exercises
8	—	**In-class project**
9	—	**In-class project**
10	—	**In-class project**
11	—	**In-class project**
12	—	**In-class project**

Assessment

Component	Format
Midterm exam	MCQ — code-based
Final exam	MCQ — code-based

Midterm coverage: first five DataCamp courses (see DataCamp courses used).

Final coverage: all midterm courses, plus Introduction to Functions in Python, Software Engineering Principles in Python, and the in-class project.

Exam rules

- **Duration:** 2 hours
- **Questions:** 40; each presents a code snippet and 5 true/false statements
- **Authorized documents:** none
- **Calculator:** not allowed
- **Draft paper:** not allowed
- **Dictionary:** not allowed

Scoring — all-or-nothing per question

Each question receives **full credit or zero**:

- Full credit if **all** true/false selections for that question are correct
- Zero credit if **any** selection is wrong or any required selection is missing
- No partial credit within a question
- No penalty scoring

Maximum score: 20 points (0.5 point per question)

Resit

Same format as the final exam. A new question set is used. Resit grade replaces the final exam grade per program rules.